Policy Optimization: Actor-Critic to PPO

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June 14, 2026

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Policy Optimization: **Recap (Day 3)**

Recap: What Day 3 Established

- The policy gradient theorem gives us:

$$\nabla_\theta \mathcal{J}(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t \geq 0} \nabla_\theta \log \pi_\theta(a_t|s_t)\, A^\pi(s_t, a_t) \right]$$

- $A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s)$ is the **advantage function** — the optimal baseline
- Using A^π instead of raw returns reduces variance without introducing bias
- **Unresolved question:** how do we estimate A^π in practice?
- Day 3 noted: we only need V^π, since the TD residual $\delta_t = r_t + \gamma V^\pi(s_{t+1}) - V^\pi(s_t)$ approximates A^π

Recap: What Today Delivers

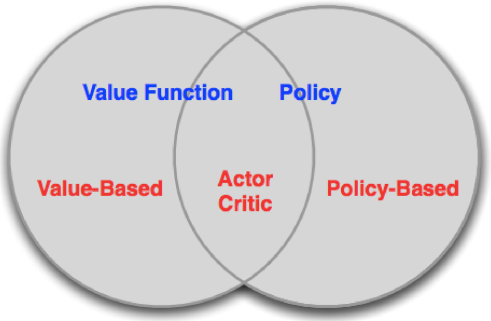
- **Today we cash that promise.**
- We learn a critic V_ϕ alongside the policy — this is the **Actor-Critic** architecture
- The critic is trained via TD learning; the TD residual δ_t serves as an online, low-variance estimate of A^π
- Then we scale up: A2C/A3C $\rightarrow$ trust region (TRPO) $\rightarrow$ clipped objective (PPO)

Learning Outcomes

- Explain why a learned critic reduces variance and enables online updates (Actor-Critic)
- Derive the TD-residual advantage estimator and connect it to the Bellman equation
- Describe A2C and A3C at a high level and identify when each is preferred
- State the TRPO trust-region constraint and why it prevents policy collapse
- Implement the PPO clipped objective and explain how clipping enforces the trust region

Policy Optimization: **Actor-Critic Methods**

The Two Worlds of RL



- **Value-based** methods (Days 1–2): learn V or Q, derive policy implicitly
- **Policy-based** methods (Day 3): learn π_θ directly, no value function
- **Actor-Critic** sits at the intersection: learn *both* a policy *and* a value function

From Baselines to Critics

- Day 3: subtracting a **state-dependent baseline** $b(s)$ reduces variance with zero bias:

$$\nabla_\theta \mathcal{J}(\theta) = \mathbb{E}_t \left[\nabla_\theta \log \pi_\theta(a_t|s_t) \left(Q^\pi(s_t, a_t) - b(s_t) \right) \right]$$

- The best choice of baseline is $b(s) = V^\pi(s)$, giving $A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$

From Baselines to Critics

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- The best choice of baseline is $b(s) = V^\pi(s)$, giving $A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$
- But we don't know V^π — we have to **learn it**
- This turns the baseline into a **critic**: a second network $V_\phi(s)$ trained alongside the actor π_θ
- **Actor** $\pi_\theta(a|s)$: selects actions, updated by policy gradient
- **Critic** $V_\phi(s)$: evaluates states, updated by TD learning

The TD Residual as Advantage

- Given a critic V_ϕ, define the **TD residual**:

$$\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$$

- When $V_\phi = V^\pi$ (a perfect critic), the **expected** TD residual equals the advantage:

$$\begin{aligned}\mathbb{E}[\delta_t \mid s_t, a_t] &= r(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1}}[V^\pi(s_{t+1})] - V^\pi(s_t) \\ &= Q^\pi(s_t, a_t) - V^\pi(s_t) \;=\; A^\pi(s_t, a_t)\end{aligned}$$

where $r(s_t, a_t) = \mathbb{E}[r_t \mid s_t, a_t]$ is the expected reward function. In practice we use the observed r_t and a learned V_ϕ, so δ_t is an approximation of A^π.

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- So we can use δ_t as a **low-variance, online** estimate of A^π — no full trajectory needed
- This is what Day 3 deferred: we now have a way to estimate A^π at every timestep
- **Critic update:** minimize the TD error squared: $\mathcal{L}(\phi) = \mathbb{E}_t\left[\delta_t^2\right]$

From n-step Returns to GAE

- Between 1-step TD and full Monte Carlo lies a whole family of **n-step advantage estimators**:

$$\hat{A}_t^{(n)} = \underbrace{r_t + \gamma r_{t+1} + \cdots + \gamma^{n-1} r_{t+n-1} + \gamma^n V_\phi(s_{t+n})}_{n\text{-step return}} - V_\phi(s_t)$$

  - Small n: low variance, high bias; large n: high variance, low bias

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- Each $\hat{A}_t^{(n)}$ telescopes into a sum of TD residuals: $\hat{A}_t^{(n)} = \sum_{l=0}^{n-1} \gamma^l \delta_{t+l}$
- **GAE** avoids picking a single n: take an **exponentially-weighted average** over all n with decay λ:

$$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = (1-\lambda) \sum_{n=1}^{\infty} \lambda^{n-1} \hat{A}_t^{(n)} = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}$$

- λ smoothly tunes the bias–variance trade-off (next slide: the limits)

GAE: The Two Limits

- GAE gives a single knob λ that interpolates the bias–variance trade-off:

$$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = \sum_{l=0}^{\infty} (\gamma\lambda)^l \, \delta_{t+l}$$

- $\lambda = 0$: pure 1-step TD — low variance, biased
- $\lambda = 1$: full Monte Carlo ($G_t - V_\phi(s_t)$) — unbiased, high variance
- $\lambda \approx 0.95$: standard in PPO implementations

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- $\lambda \approx 0.95$: standard in PPO implementations
- We write $\hat{A}_t$ for short: a **practical estimate** of the true advantage $A^\pi(s_t, a_t)$ from Day 1
- **Used by PPO in the lab** — we revisit this when implementing the algorithm

Actor-Critic Algorithm

Initialise actor θ, critic ϕ;
for *each episode* **do**
  observe initial state s_0;
  for $t = 0, 1, 2, \ldots$ **do**
    sample $a_t \sim \pi_\theta(\cdot|s_t)$;
    take a_t, observe r_t, s_{t+1};
    $\delta_t \leftarrow r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$;
    $\theta \leftarrow \theta + \alpha_\theta \nabla_\theta \log \pi_\theta(a_t|s_t)\, \delta_t$;
    $\phi \leftarrow \phi - \alpha_\phi\, \delta_t \nabla_\phi V_\phi(s_t)$;
  end
end

Algorithm: One-step Actor-Critic (online)

- **Actor** steps along the policy gradient, weighted by δ_t
- **Critic** regresses $V_\phi(s_t)$ toward $r_t + \gamma V_\phi(s_{t+1})$
- The same δ_t drives *both* updates — no full trajectory needed
- This is the pure **TD** form; batched A2C/PPO swap the single δ_t for a **GAE** estimate $\hat{A}_t$ over minibatches

Why Actor-Critic Helps

- **Variance reduction vs REINFORCE:** the TD residual δ_t is a much lower-variance signal than $G_t - b$ because it bootstraps off V_ϕ rather than waiting for the full trajectory
- **Online updates:** no need to wait until the end of an episode — the critic provides a training signal at every timestep
- **Bootstrapping:** propagating value estimates forward allows the agent to learn from partial trajectories and even in continuing (non-episodic) tasks

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- **Bootstrapping:** propagating value estimates forward allows the agent to learn from partial trajectories and even in continuing (non-episodic) tasks
- **Trade-off:** if V_ϕ is a bad critic, the TD residual is a biased advantage estimate — the actor is being guided by a noisy critic
- Both networks must be trained together carefully; instability can arise if the critic lags the actor
- This is the bias-variance trade-off formalised by GAE

A2C: Synchronous Advantage Actor-Critic

- **Key idea:** run N parallel actors sharing the same θ and ϕ
- All actors step simultaneously; gradients are aggregated and a single synchronous update is applied
- The shared critic sees diverse experience from all environments — more stable training signal
- **Policy loss:** $\mathcal{L}^{\pi}(\theta) = -\mathbb{E}_t\left[\log \pi_\theta(a_t|s_t)\,\hat{A}_t\right]$
- **Value loss:** $\mathcal{L}^{V}(\phi) = \mathbb{E}_t\left[\delta_t^2\right]$
- A2C is what most modern Actor-Critic implementations actually use
- Parallel actors de-correlate experience without a replay buffer (on-policy)

A3C: Asynchronous Variant

- **A3C** (Mnih et al., 2016): Asynchronous Advantage Actor-Critic
- Workers run asynchronously on CPU threads, applying gradients to a shared model without locks (Hogwild-style updates)
- The asynchrony acts as an implicit decorrelation mechanism — each worker is at a different point in its trajectory

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- A3C's asynchrony introduces **gradient staleness**: a worker may apply a gradient computed under an older version of θ

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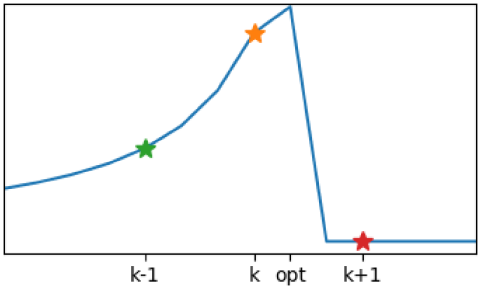
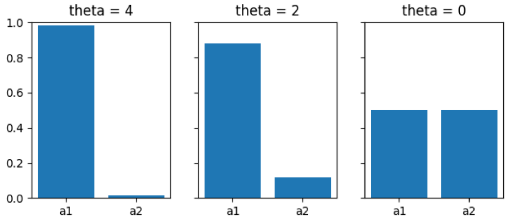
- A3C's asynchrony introduces **gradient staleness**: a worker may apply a gradient computed under an older version of θ
- **Going forward:** we use A2C as the basis for PPO; A3C is historical context
- But parallel actors only *accelerate* data collection — they do not fix the underlying issue that **a single gradient step can still destroy the policy**. That motivates the next section.

Why Step Size Matters





- A large gradient step in *parameter space* can cause a catastrophically large shift in *policy space*
- Once a policy collapses (e.g., always picks one bad action), the agent collects bad data and the policy is hard to recover
- We need to control how much the policy distribution changes, not just how much θ changes

Surrogate Objective & Importance Sampling

- **Relative policy performance identity:** the improvement of π_θ over $\pi_{\theta_{\text{old}}}$ can be written as:

$$\mathcal{J}(\theta) - \mathcal{J}(\theta_{\text{old}}) = \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t\geq 0}\gamma^t A^{\pi_{\theta_{\text{old}}}}(s_t, a_t)\right]$$

- Since we collect data under $\pi_{\theta_{\text{old}}}$, re-weight via importance sampling. Define the probability ratio:[1]

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$$

- This gives the **surrogate objective**:

$$\mathcal{L}(\theta) = \mathbb{E}_t[r_t(\theta)\, A^{\pi_{\theta_{\text{old}}}}(s_t, a_t)]$$

[1] $r_t(\theta)$ here is the probability ratio, not the reward r_t from earlier lectures — distinguished by the θ argument.

Surrogate Objective: Why It Needs a Constraint

- **Problem:** if π_θ drifts far from $\pi_{\theta_{\text{old}}}$, the ratio $r_t(\theta)$ explodes and the estimate becomes unreliable
- **Solution: constrain how far π_θ can move** from $\pi_{\theta_{\text{old}}}$
- Next we make this precise: *why* the surrogate is principled in the first place, and *why* importance sampling specifically demands a constraint — before turning to TRPO and PPO, which both implement the constraint in different ways

Foundations: Performance Difference Lemma

- The identity on the previous slide has a name: the **Performance Difference Lemma**[1]

$$\mathcal{J}(\theta) - \mathcal{J}(\theta_{\text{old}}) = \mathbb{E}_{\tau \sim \pi_\theta}\left[\sum_{t \geq 0} \gamma^t A^{\pi_{\theta_{\text{old}}}}(s_t, a_t)\right]$$

- **Why this matters:** it makes the surrogate $\mathcal{L}(\theta)$ a **principled approximation** to the true objective $\mathcal{J}(\theta)$ — not a heuristic
- When $\pi_\theta \approx \pi_{\theta_{\text{old}}}$, the state distributions induced by the two policies are similar, so we can substitute samples from $\pi_{\theta_{\text{old}}}$ for samples from π_θ with controlled error
- As π_θ drifts away from $\pi_{\theta_{\text{old}}}$, the approximation degrades. This is the theoretical reason we need a trust region — not just empirical observation that "big steps break things"

[1] Lemma originally due to Kakade & Langford (2002); foundation of TRPO in Schulman et al. (2015).

Foundations: Importance Sampling Variance

- Using data from $\pi_{\theta_{\text{old}}}$ requires importance sampling. The IS-weighted estimator has variance

$$\text{Var}\left[\frac{P(x)}{Q(x)}\, f(x)\right] = \mathbb{E}_{x\sim P}\left[\frac{P(x)}{Q(x)}\, f(x)^2\right] - \mathbb{E}[f]^2$$

- The variance **explodes** when the ratio $\frac{P}{Q}$ is large where f^2 is large — a single bad sample dominates the estimate
- **Concretely:** if $\pi_{\theta_{\text{old}}}(a|s)$ is tiny where $\pi_\theta(a|s)$ is large, the ratio $r_t(\theta)$ blows up. One rare sample under the old policy gets enormous weight and swings the entire gradient estimate

[2] Full derivation in OpenAI's Spinning Up: TRPO and Schulman et al. (2015).

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- For **trajectory-level** IS: $\frac{p(\tau|\pi_\theta)}{p(\tau|\pi_{\theta_{\text{old}}})} = \prod_{t=0}^{T} r_t(\theta)$. Small per-step differences **compound multiplicatively** over T steps — the variance is hopeless
- TRPO and PPO use **single-step ratios** and constrain them to stay near 1 — introduced next[2]

[2]Full derivation in OpenAI's Spinning Up: TRPO and Schulman et al. (2015).

Policy Optimization: **TRPO**

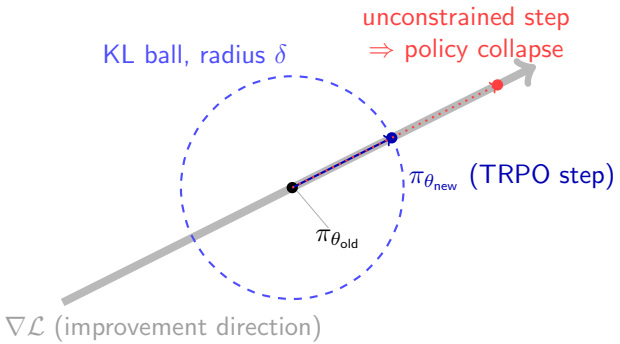
TRPO: KL Trust Region

- TRPO formalises the constraint: maximize the surrogate objective subject to a KL-divergence bound:

$$\max_{\theta} \mathcal{L}(\theta) \quad \text{s.t.} \quad \mathbb{E}_t[D_{KL}(\pi_{\theta_{\text{old}}}(\cdot|s_t) \,\|\, \pi_\theta(\cdot|s_t))] \leq \delta$$

- Keeps the new policy inside a **trust region** in policy space, not parameter space
- Solved with conjugate gradient + backtracking line search (second-order): largest natural-gradient step that still satisfies the KL constraint
- **Cost:** needs the Fisher information matrix — expensive and hard to scale. *This is exactly what motivates PPO.*

TRPO: Trust Region in Policy Space



- Each update stays within a **KL ball** around $\pi_{\theta_{\text{old}}}$ (*policy* space, not parameter space): the step climbs $\mathcal{L}$ but is clipped at the boundary, preventing the overshoot that collapses the policy

Policy Optimization: **PPO**

PPO: First-Order Trust Region

- TRPO achieves monotonic improvement but requires second-order optimisation (conjugate gradient, Fisher matrix) — expensive and complex to implement
- **Question:** can we get TRPO-level stability with plain SGD?

PPO: First-Order Trust Region

- TRPO achieves monotonic improvement but requires second-order optimisation (conjugate gradient, Fisher matrix) — expensive and complex to implement
- **Question:** can we get TRPO-level stability with plain SGD?
- **PPO** (Schulman et al., 2017) proposes two variants:
  - **PPO-KL:** add a KL penalty to the objective and adapt the penalty coefficient β over training
  - **PPO-Clip:** clip the probability ratio $r_t(\theta)$ to prevent large policy updates
- PPO-Clip is the standard variant — simpler, no hyperparameter tuning on β, works at least as well empirically

PPO Clipped Objective

- Recall the probability ratio: $r_t(\theta) = \dfrac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$
- The **PPO-Clip objective** is:

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(r_t(\theta)\,\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\,\hat{A}_t\Big)\Big]$$

- ε is a small hyperparameter (typically $\varepsilon = 0.2$)

PPO Clipped Objective

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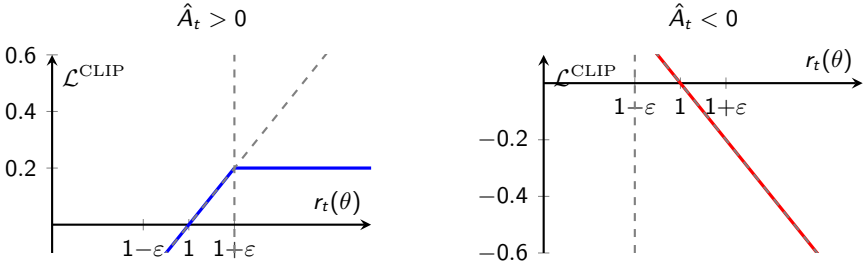
- ε is a small hyperparameter (typically $\varepsilon = 0.2$)
- **Why clip + min?**
  - When $\hat{A}_t > 0$: clipping caps r_t at $1+\varepsilon$, removing the incentive to increase the ratio further
  - When $\hat{A}_t < 0$: clipping floors r_t at $1-\varepsilon$, removing the incentive to decrease the ratio further
  - The **min** ensures the objective is a lower bound — we never benefit from going outside $[1-\varepsilon, 1+\varepsilon]$
- This enforces a soft trust region without any second-order computation

Visualising the Clip



- **Left ($\hat{A}_t > 0$):** benefit grows linearly with r_t until $1+\varepsilon$, then is clipped flat — no reward for pushing the policy further
- **Right ($\hat{A}_t < 0$):** penalty grows as r_t decreases until $1-\varepsilon$, then is clipped flat
- Gray dashed line = unclipped surrogate; colored solid = clipped (what PPO optimises)

PPO Full Algorithm

Algorithm 5 PPO with Clipped Objective

Input: initial policy parameters θ_0, clipping threshold ϵ
for $k = 0, 1, 2, \ldots$ **do**
  Collect set of partial trajectories $\mathcal{D}_k$ on policy $\pi_k = \pi(\theta_k)$
  Estimate advantages $\hat{A}_t^{\pi_k}$ using any advantage estimation algorithm
  Compute policy update

$$\theta_{k+1} = \arg\max_{\theta} \mathcal{L}_{\theta_k}^{CLIP}(\theta)$$

  by taking K steps of minibatch SGD (via Adam), where

$$\mathcal{L}_{\theta_k}^{CLIP}(\theta) = \mathop{\mathrm{E}}_{\tau \sim \pi_k} \left[\sum_{t=0}^{T} \left[\min(r_t(\theta)\hat{A}_t^{\pi_k}, \text{clip}\left(r_t(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_t^{\pi_k}) \right] \right]$$

end for

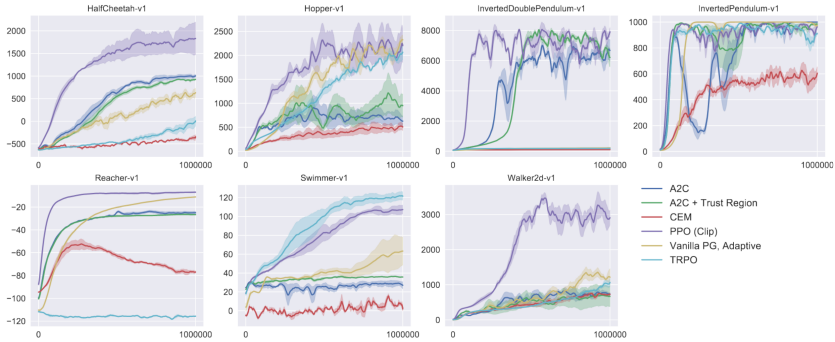
- **Sample efficiency:** PPO runs multiple SGD epochs over the same rollout — unlike A2C which discards data after one update
- **Combined loss:** $\mathcal{L}(\theta, \phi) = \mathcal{L}^{\text{CLIP}}(\theta) - c_1\mathcal{L}^V(\phi) + c_2\mathcal{H}(\pi_\theta)$ (entropy bonus encourages exploration)
- GAE is used to compute $\hat{A}_t$; the entropy term will reappear in SAC (Day 6)

PPO in Practice

- The clipped objective is only half the story — these implementation details matter *as much as* ε, and explain why two "correct" PPO implementations can differ by an order of magnitude[2]
- **Advantage normalisation:** standardise $\hat{A}_t$ per minibatch (stabilises gradient scale)
- **Value-loss clipping:** clip the critic update like the policy
- **Gradient clipping:** bound global grad-norm (typically 0.5)
- **KL early stopping:** end the epoch loop once mean KL exceeds a target — a cheap trust-region safeguard
- **Obs / reward normalisation:** running normalisation keeps inputs and returns well-scaled
- **Orthogonal init:** small policy-head gain avoids an over-confident initial policy

[2] Andrychowicz et al. (2020), *What Matters in On-Policy RL?*

PPO Empirical Performance



- PPO matches or exceeds TRPO with far less complexity; outperforms vanilla PG and CEM on most benchmarks
- This is the algorithm you will implement in the lab[3]

[3] Schulman, Wolski, Dhariwal, Radford, Klimov, 2017

Policy Optimization: **GRPO**

PPO Without a Critic: GRPO

- The fragile part of the Actor-Critic loop is the **critic** V_ϕ — training two networks together is the main source of instability
- **Group Relative Policy Optimization** (GRPO; Shao et al., 2024) removes the critic entirely

PPO Without a Critic: GRPO

- The fragile part of the Actor-Critic loop is the **critic** V_ϕ — training two networks together is the main source of instability
- **Group Relative Policy Optimization** (GRPO; Shao et al., 2024) removes the critic entirely
- Per prompt/state, sample a **group** of G outputs under $\pi_{\theta_{\text{old}}}$, scored with rewards R_i
- Use the **group mean as the baseline** — a Monte Carlo advantage, no value function:

$$\hat{A}_i = \frac{R_i - \overbrace{\text{mean}(R_1, \ldots, R_G)}^{\text{the “Group Relative” baseline}}}{\underbrace{\text{std}(R_1, \ldots, R_G)}_{\text{per-group advantage normalisation}}}$$

GRPO: What Is and Isn't the Baseline

- Only the **mean-subtraction** is the GRPO contribution — that is the "Group Relative" baseline replacing the critic V_ϕ
- The **std-normalisation** is a *separate* stabilisation trick: it is exactly the per-minibatch advantage normalisation from the PPO-in-Practice slide, applied per group — not part of the baseline
- So a Monte Carlo baseline does *not* "need" variance correction; the std term is an orthogonal, optional stabiliser

GRPO: What Is and Isn't the Baseline

- Only the **mean-subtraction** is the GRPO contribution — that is the "Group Relative" baseline replacing the critic V_ϕ
- The **std-normalisation** is a *separate* stabilisation trick: it is exactly the per-minibatch advantage normalisation from the PPO-in-Practice slide, applied per group — not part of the baseline
- So a Monte Carlo baseline does *not* "need" variance correction; the std term is an orthogonal, optional stabiliser
- Plug $\hat{A}_i$ into the *same* PPO-Clip objective (plus a KL term to a reference policy)
- **Why it matters:** no critic $\Rightarrow$ no two-network instability and $\approx$half the memory
- Since 2024 GRPO is the default for **LLM RL** (DeepSeek-R1, many open recipes) — revisited in **Day 9 (RLHF)**, where the reward is a learned preference model

Policy Optimization: **Summary**

The Algorithm Family Tree

REINFORCE $\rightarrow$ +baseline $\rightarrow$ +critic (A2C) $\rightarrow$ +GAE $\rightarrow$ +trust region (TRPO) $\rightarrow$ +clipping (PPO) $\rightarrow$ $-$critic (GRPO)

Algorithm	**On-policy?**	**Variance**	**Sample eff.**	**Complexity**
REINFORCE	Yes	High	Low	Simple
A2C	Yes	Medium	Medium	Moderate
TRPO	Yes	Medium	Medium	High
PPO	Yes	Medium	Medium	Low
GRPO	Yes	Medium	Medium	Low (no critic)

Summary

- A **critic** V_ϕ learns to estimate state values via TD; the TD residual δ_t approximates A^π cheaply
- **A2C** parallelises data collection with N synchronous actors; A3C uses asynchronous workers (empirically A2C $\geq$ A3C)
- **TRPO** guarantees monotonic improvement via a KL trust region but is expensive to compute
- **PPO-Clip** approximates the trust region with a first-order method: clip $r_t(\theta)$ to $[1-\varepsilon, 1+\varepsilon]$ and take the min with the unclipped objective
- PPO is the workhorse of modern on-policy RL: simple, stable, widely used in robotics, games, and RLHF
- **GRPO** drops the critic and uses group-mean baselines — now the default for LLM RL (DeepSeek-R1)

Limitations of Actor-Critic & PPO

- **Function approximation bias:** the TD residual δ_t is only an unbiased advantage estimate if $V_\phi = V^\pi$. An inaccurate critic introduces bias into the policy gradient
- **Two-network instability:** if the critic lags behind the actor (or vice versa), the feedback loop can diverge — both networks must be trained carefully together (GRPO sidesteps this by dropping the critic entirely)
- **Hyperparameter sensitivity:** PPO performance is sensitive to ε (clip range), λ (GAE), number of epochs per rollout, and entropy coefficient c_2
- **On-policy sample inefficiency:** all methods in this lecture discard experience after each update. In complex environments this can require billions of environment steps
- **Dense-reward assumption:** variance reduction via advantage estimation works well when rewards are frequent. Sparse-reward tasks remain challenging for all on-policy methods

What's Next

- **Key limitation:** on-policy methods discard all experience after each update — data-hungry for complex environments

What's Next

- **Key limitation:** on-policy methods discard all experience after each update — data-hungry for complex environments
- **Day 5 — DDPG & TD3:** off-policy deterministic policies for continuous control — replay buffer, action-value critic $Q_\phi(s, a)$, gradient through $Q_\phi(s, \mu_\theta(s))$
- **Day 6 — SAC:** max-entropy Actor-Critic — the same $c_2\mathcal{H}(\pi_\theta)$ bonus you saw in PPO, now **central** to the objective
- **Day 7 — Exploration:** that entropy bonus *is* exploration via a stochastic policy; Day 7 makes exploration the explicit objective
- **Day 9 — RLHF:** PPO is the algorithm underneath the original InstructGPT/ChatGPT pipeline; GRPO is its critic-free successor

Policy Optimization: **References**

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Backup — TRPO Algorithm (pseudocode)

Algorithm 3 Trust Region Policy Optimization

Input: initial policy parameters θ_0
for $k = 0, 1, 2, \ldots$ **do**
Collect set of trajectories $\mathcal{D}_k$ on policy $\pi_k = \pi(\theta_k)$
Estimate advantages $\hat{A}_t^{\pi_k}$ using any advantage estimation algorithm
Form sample estimates for

- policy gradient $\hat{g}_k$ (using advantage estimates)
- and KL-divergence Hessian-vector product function $f(v) = \hat{H}_k v$

Use CG with n_{cg} iterations to obtain $x_k \approx \hat{H}_k^{-1} \hat{g}_k$
Estimate proposed step $\Delta_k \approx \sqrt{\frac{2\delta}{x_k^T \hat{H}_k x_k}} x_k$
Perform backtracking line search with exponential decay to obtain final update

$$\theta_{k+1} = \theta_k + \alpha^j \Delta_k$$

end for

- Reference only — students implement PPO, not TRPO, in the lab (see Schulman et al., 2015)